

SQL Functions Assignment

Question 1: Create a ranking of students based on score (highest first).

Answer: The SQL Query we used to completed is given Task is :

SELECT

```
RANK() OVER (ORDER BY score DESC) AS student_rank,  
student_id,  
name,  
course,  
score,  
attendance,  
mentor,  
city
```

FROM Student_Performance;

The Result Grid:

	student_rank	student_id	name	course	score	attendance	mentor	city
▶	1	107	Arjun Verma	Tableau	95	98	Ms. Kapoor	Delhi
	2	114	Ishita Joshi	Data Science	93	97	Dr. Sharma	Bengaluru
	3	110	Priya Desai	SQL	92	94	Mr. Iyer	Bengaluru
	4	103	Kabir Khanna	Python	91	96	Ms. Nair	Mumbai
	5	113	Rohan Gupta	SQL	89	91	Mr. Iyer	Delhi
	6	101	Aarav Mehta	Data Science	88	92	Dr. Sharma	Mumbai
	7	111	Siddharth Jain	Python	85	90	Ms. Nair	Mumbai
	8	104	Tanvi Patel	SQL	84	89	Mr. Iyer	Bengaluru
	9	108	Meera Pillai	Tableau	82	87	Ms. Kapoor	Kochi
	10	109	Nikhil Rao	Data Science	79	82	Dr. Sharma	Chennai
	11	102	Riva Singh	Data Science	76	85	Dr. Sharma	Delhi

Question 2: Show each student's score and the previous student's score (based on score order).

Answer: The SQL Query we used to completed is given Task is:

```
SELECT
    student_id,
    name,
    course,
    score,
    LAG(score) OVER (ORDER BY score DESC) AS previous_student_score
FROM Student_Performance;
```

The Result Grid:

Result Grid

Filter Rows:

Export:

Wrap Cell Conte

	student_id	name	course	score	previous_student_score
▶	107	Arjun Verma	Tableau	95	NULL
	114	Ishita Joshi	Data Science	93	95
	110	Priya Desai	SQL	92	93
	103	Kabir Khanna	Python	91	92
	113	Rohan Gupta	SQL	89	91
	101	Aarav Mehta	Data Science	88	89
	111	Siddharth Jain	Python	85	88
	104	Tanvi Patel	SQL	84	85
	108	Meera Pillai	Tableau	82	84
	109	Nikhil Rao	Data Science	79	82
	102	Riva Singh	Data Science	76	79

Question 3: Convert all student names to uppercase and extract the month name from join_date.

Answer: The SQL Query we used to completed is given Task is:

```
SELECT
    student_id,
    UPPER(name) AS student_name,
    course,
    score,
```

join_date,

MONTHNAME(join_date) AS join_month

FROM Student_Performance;

The Result Grid:

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	student_id	student_name	course	score	join_date	join_month
▶	101	AARAV MEHTA	Data Science	88	2023-06-12	June
	102	RIYA SINGH	Data Science	76	2023-07-01	July
	103	KABIR KHANNA	Python	91	2023-06-20	June
	104	TANVI PATEL	SQL	84	2023-05-30	May
	105	AYESHA KHAN	Python	67	2023-07-10	July
	106	DEV SHARMA	SQL	73	2023-05-28	May
	107	ARJUN VERMA	Tableau	95	2023-06-15	June
	108	MEERA PILLAI	Tableau	82	2023-06-18	June
	109	NIKHIL RAO	Data Science	79	2023-07-05	July
	110	PRIYA DESAI	SQL	92	2023-05-27	May
	111	SIDDHARTH J...	Pvthon	85	2023-07-02	Julv

Question 4: Show each student's name and the next student's attendance (ordered by attendance).

Answer: The SQL Query we used to completed is given Task is:

SELECT

student_id,

name,

attendance,

LEAD(attendance) OVER (ORDER BY attendance DESC) AS next_student_attendance

FROM Student_Performance;

The Result Grid:

student_id	name	attendance	next_student_attendance
107	Arjun Verma	98	97
114	Ishita Joshi	97	96
103	Kabir Khanna	96	94
110	Priya Desai	94	92
101	Aarav Mehta	92	91 91
113	Rohan Gupta	91	90
111	Siddharth Jain	90	89
104	Tanvi Patel	89	87
108	Meera Pillai	87	85
102	Riya Singh	85	84
115	Yuvrai Rao	84	83

Question 5: Assign students into 4 performance groups using NTILE().

Answer: The SQL Query we used to completed is given Task is:

```
SELECT
    student_id,
    name,
    course,
    score,
    NTILE(4) OVER (ORDER BY score DESC) AS performance_group,
    CASE NTILE(4) OVER (ORDER BY score DESC)
        WHEN 1 THEN 'Excellent'
        WHEN 2 THEN 'Good'
        WHEN 3 THEN 'Average'
        WHEN 4 THEN 'Needs Improvement'
    END AS performance_label
FROM Student_Performance;
```

The Result Grid:

	student_id	name	course	score	performance_group	performance_label
▶	107	Arjun Verma	Tableau	95	1	Excellent
	114	Ishita Joshi	Data Science	93	1	Excellent
	110	Priya Desai	SQL	92	1	Excellent
	103	Kabir Khanna	Python	91	1	Excellent
	113	Rohan Gupta	SQL	89	2	Good
	101	Aarav Mehta	Data Science	88	2	Good
	111	Siddharth Jain	Python	85	2	Good
	104	Tanvi Patel	SQL	84	2	Good
	108	Meera Pillai	Tableau	82	3	Average
	109	Nikhil Rao	Data Science	79	3	Average
	102	Riva Singh	Data Science	76	3	Average

Question 6: For each course, assign a row number based on attendance (highest first).

Answer: The SQL Query we used to completed is given Task is:

```
SELECT
    student_id,
    name,
    course,
    attendance,
    ROW_NUMBER() OVER (PARTITION BY course ORDER BY attendance DESC) AS
row_num
FROM Student_Performance;
```

The Result Grid is:

	student_id	name	course	attendance	row_num
▶	114	Ishita Joshi	Data Science	97	1
	101	Aarav Mehta	Data Science	92	2
	102	Riya Singh	Data Science	85	3
	109	Nikhil Rao	Data Science	82	4
	103	Kabir Khanna	Python	96	1
	111	Siddharth Jain	Python	90	2
	115	Yuvraj Rao	Python	84	3
	105	Ayesha Khan	Python	81	4
	110	Priya Desai	SQL	94	1
	113	Rohan Gupta	SQL	91	2
	104	Tanvi Patel	SOL	89	3

Question 7: Calculate the number of days each student has been enrolled (from join_date to today). (Assume current date = '2025-01-01')

Answer: The SQL Query we used to completed is given Task is:

```
SELECT
    student_id,
    name,
    course,
    join_date,
    DATEDIFF('2025-01-01', join_date) AS days_enrolled
```

FROM Student_Performance;

The Result Grid is:

	student_id	name	course	join_date	days_enrolled
▶	101	Aarav Mehta	Data Science	2023-06-12	569
	102	Riya Singh	Data Science	2023-07-01	550
	103	Kabir Khanna	Python	2023-06-20	561
	104	Tanvi Patel	SQL	2023-05-30	582
	105	Ayesha Khan	Python	2023-07-10	541
	106	Dev Sharma	SQL	2023-05-28	584
	107	Arjun Verma	Tableau	2023-06-15	566
	108	Meera Pillai	Tableau	2023-06-18	563
	109	Nikhil Rao	Data Science	2023-07-05	546
	110	Priya Desai	SQL	2023-05-27	585
	111	Siddharth Jain	Pvthon	2023-07-02	549

Question 8: Format join_date as “Month Year” (e.g., “June 2023”).

Answer: The SQL Query we used to completed is given Task is:

SELECT

student_id,

name,

course,

join_date,

DATE_FORMAT(join_date, '%M %Y') AS join_month_year

FROM Student_Performance;

The Result Grid is:

	student_id	name	course	join_date	join_month_year
▶	101	Aarav Mehta	Data Science	2023-06-12	June 2023
	102	Riya Singh	Data Science	2023-07-01	July 2023
	103	Kabir Khanna	Python	2023-06-20	June 2023
	104	Tanvi Patel	SQL	2023-05-30	May 2023
	105	Ayesha Khan	Python	2023-07-10	July 2023
	106	Dev Sharma	SQL	2023-05-28	May 2023
	107	Arjun Verma	Tableau	2023-06-15	June 2023
	108	Meera Pillai	Tableau	2023-06-18	June 2023
	109	Nikhil Rao	Data Science	2023-07-05	July 2023
	110	Priya Desai	SQL	Data Science	May 2023
	111	Siddharth Jain	Pvthon	2023-07-02	Julv 2023

Question 9: Replace the city 'Mumbai' with 'MUM' for display purposes.

Answer: The SQL Query we used to completed is given Task is:

```
SELECT
    student_id,
    name,
    course,
    score,
    REPLACE(city, 'Mumbai', 'MUM') AS city
FROM Student_Performance;
```

The Result Grid is:

	student_id	name	course	score	city
▶	101	Aarav Mehta	Data Science	88	MUM
	102	Riya Singh	Data Science	76	Delhi
	103	Kabir Khanna	Python	91	MUM
	104	Tanvi Patel	SQL	84	Bengaluru
	105	Ayesha Khan	Python	67	Hyderabad
	106	Dev Sharma	SQL	73	Pune
	107	Arjun Verma	Tableau	95	Delhi
	108	Meera Pillai	Tableau	82	Kochi
	109	Nikhil Rao	Data Science	79	Chennai
	110	Priya Desai	SQL	92	Bengaluru
	111	Siddharth Jain	Pvthon	85	MUM

Question 10: For each course, find the highest score using FIRST_VALUE().

Answer: The SQL Query we used to completed is given Task is:

```
SELECT
    student_id,
    name,
    course,
    score,
    FIRST_VALUE(score) OVER (PARTITION BY course ORDER BY score DESC) AS
    highest_score
FROM Student_Performance;
```

The Result Grid is:

	student_id	name	course	score	highest_score
▶	114	Ishita Joshi	Data Science	93	93
	101	Aarav Mehta	Data Science	88	93
	109	Nikhil Rao	Data Science	79	93
	102	Riya Singh	Data Science	76	93
	103	Kabir Khanna	Python	91	91
	111	Siddharth Jain	Python	85	91
	115	Yuvraj Rao	Python	71	91
	105	Ayesha Khan	Python	67	91
	110	Priya Desai	SQL	92	92
	113	Rohan Gupta	SQL	89	92